

RESEARCH ARTICLE

10.1002/2016WR020258

Key Points:

- Vertical preferential flow paths in tropical catchments have a dominant effect on groundwater recharge and base flow
- Near-surface lateral preferential flow paths are the main contributor to the runoff peak flow characteristics
- Model results are corroborated by geochemical and geophysical observations to avoid calibration

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Citation:

Cheng, Y., F. L. Ogden, and J. Zhu (2017), Earthworms and tree roots: A model study of the effect of preferential flow paths on runoff generation and groundwater recharge in steep, saprolitic, tropical lowland catchments, *Water Resour. Res.*, 53, 5400–5419, doi:10.1002/2016WR020258.

Received 13 DEC 2016

Accepted 4 JUN 2017

Accepted article online 14 JUN 2017

Published online 4 JUL 2017

Earthworms and tree roots: A model study of the effect of preferential flow paths on runoff generation and groundwater recharge in steep, saprolitic, tropical lowland catchments

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Abstract Preferential flow paths (PFPs) affect the hydrological response of humid tropical catchments but have not received sufficient attention. We consider PFPs created by tree roots and earthworms in a near-surface soil layer in steep, humid, tropical lowland catchments and hypothesize that observed hydrological behaviors can be better captured by reasonably considering PFPs in this layer. We test this hypothesis by evaluating the performance of four different physically based distributed model structures without and with PFPs in different configurations. Model structures are tested both quantitatively and qualitatively using hydrological, geophysical, and geochemical data both from the Smithsonian Tropical Research Institute Agua Salud Project experimental catchment(s) in Central Panama and other sources in the literature. The performance of different model structures is evaluated using runoff Volume Error and three Nash-Sutcliffe efficiency measures against observed total runoff, stormflows, and base flows along with visual comparison of simulated and observed hydrographs. Two of the four proposed model structures which include both lateral and vertical PFPs are plausible, but the one with explicit simulation of PFPs performs the best. A small number of vertical PFPs that fully extend below the root zone allow the model to reasonably simulate deep groundwater recharge, which plays a crucial role in base flow generation. Results also show that the shallow lateral PFPs are the main contributor to the observed high flow characteristics. Their number and size distribution are found to be more important than the depth distribution. Our model results are corroborated by geochemical and geophysical observations.

1. Introduction

Hydrological processes in humid tropical lowlands are significantly different from those common in temperate climates [Mosley, 1979; Bonell, 1993; Sidle *et al.*, 2000, 2001; Uchida *et al.*, 2001; Bonell and Bruijnzeel, 2005]. These differences arise primarily because the tropics experience consistent high temperatures that allow for continuous and intense biological activities and competitions [Bonell and Bruijnzeel, 2005; Wohl *et al.*, 2012]. These factors increase the rate at which bedrock is weathered. In regions with igneous, metamorphic, or sedimentary bedrock, this leads to the formation of deeper, saprolitic soils, while intense biological activity in these soils creates preferential flow paths (PFPs) by altering the geometric properties of the soil and leaving texture unaltered [Bachmair *et al.*, 2009]. These PFPs contribute significantly to catchment hydrological behavior by affecting the partitioning of rainfall into different hydrological pathways [Jones, 2010] but have not received sufficient attention [Beven and Germann, 1982, 2013], particularly in the context of hydrological modeling. The role of macropores in both rapid runoff [Noguchi *et al.*, 1999; Kinner and Stalard, 2004] and groundwater recharge [Cuthbert and Tindimugaya, 2010] is a fundamental research question [Beven and Germann, 1982] that has yet to be fully addressed within humid tropical hillslopes and basins [Chappell, 2010].

Many field studies have examined potential sources and impact factors of preferential flow. Hölscher *et al.* [1998] and Ghimire *et al.* [2017] demonstrated that stemflow is a possible contributor to macropore flow as it concentrated at the base of tree stems where root channels typically start. Soil types [Elsenbeer, 2001; Chappell, 2010], bedrock permeability [Onda *et al.*, 2001], and threshold soil moisture values [Tromp-Van Meerveld and McDonnell, 2006a, 2006b; Fu *et al.*, 2013a, 2013b; Du *et al.*, 2016] all have a large impact on

runoff generation mechanisms across different soils in the humid tropics as they define the width of a spectrum of possible hillslope hydrologic flow paths [Elsenbeer, 2001; Chappell, 2010].

Field experiments have also accomplished the subsurface flow path visualization. Kurtz *et al.* [2011] for the first time used Ge and Si as tracers to study hydrological pathways to the stream during storm events in Puerto Rico. Bachmair *et al.* [2009] and Zwartendijk *et al.* [2017] conducted blue dye experiments to characterize flow patterns. Schellekens *et al.* [2004] used a combination of saturated hydrological conductivity (K_{sat}) profiles, chemical tracers, and shallow geophysics to investigate various flow paths within the Bisley forest in Puerto Rico. The findings have consistently identified shallow lateral preferential flow paths (LPFPs) in tropical forest catchments [Elsenbeer and Cassel, 1990; Elsenbeer, 2001; Schellekens *et al.*, 2004; Ziegler *et al.*, 2004; Gardner, 2015; Zwartendijk *et al.*, 2017]. These LPFPs are mainly formed from living and decayed horizontally orientated tree roots [van Noordwijk *et al.*, 1991; Noguchi *et al.*, 1997, 1999; Sidle *et al.*, 2001; Bachmair *et al.*, 2009; Zwartendijk *et al.*, 2017]. The LPFPs can result in very different bulk soil hydraulic properties between upper soil layer and lower soil layers [Noguchi *et al.*, 1997; Kurtz *et al.*, 2011]. Their impact on both streamflow generation [Elsenbeer and Cassel, 1990; Elsenbeer, 2001] and rapid runoff responses [Sidle *et al.*, 2001; Weiler and McDonnell, 2007] was well documented; but models that describe these effects are still needed.

Studies have also shown that deep penetration of water into the tropical soils occurs through preferential flow in vertically oriented macropores, mainly earthworm channels [Zou and Gonzalez, 1997; Blanchart *et al.*, 1999; Bachmair *et al.*, 2009; Colloff *et al.*, 2010]. Earthworm burrowing promotes good structure and aeration of soils and creates vertical preferential flow paths (VPFPs) with high continuity and low tortuosity, resulting in good penetration of water [Aubertin, 1971; Gjettermann *et al.*, 1997; Bachmair *et al.*, 2009]. Studies showed that the VPFPs allow rain water to bypass the soil matrix [Edwards *et al.*, 1988; Munyankusi *et al.*, 1994; Meng *et al.*, 2016] and propagate below the root zone where it is available to become groundwater recharge. Larsen *et al.* [2012] examined the effects of earthworm burrowing on overland flow at plot scale by applying an exclusion strategy in eastern Puerto Rico and found overland flow increased approximately twofold resulting from the exclusion of earthworms.

It is widely recognized that in tropical soil environments water flow through these PFPs could not be modeled using texture-related properties alone [Hendrickx and Flury, 2001]. Schellekens [2000] performed modeling studies on water yield and runoff response in a tropical rain forest catchment and concluded that the absence of a separate macropore flow component in the models limited the predictive capability of the model. Gieritz *et al.* [2006] also found that the neglecting of macropores resulted in a poorer representation of the processes in a tropical headwater catchment and an integration of macropore flow should ameliorate the model performance. The lack of explicit representation of preferential flow in the model resulted in the tradeoffs between accurate simulation of peak and base flows [Beckers and Alila, 2004; Cuartas *et al.*, 2012]. But because of the inadequacy in describing PFPs quantitatively [Léonard and Rajot, 2001; Chappell, 2010], explicit simulation of preferential flow paths and accurate process-based modeling remains a challenge in humid tropical environments. Another common problem of complex physically based models at hillslope scale in simulating preferential flow is overparameterization and subsequent difficulties with parameter identifiability [Beven and Germann, 2013]. Strategies are required to avoid the so-called “calibration-trap” that leads to charges of model overparameterization [Beven, 2006], and help to demonstrate that models that are called “overparameterized” might also be considered “data starved.”

Recent advances in soil imaging techniques enable not only visualization but also quantification of various macropore structure characteristics from pore to catchment scales [Jarvis *et al.*, 2016]. The increasing availability in observation techniques has created new opportunities for preferential flow modeling. In this study, we took advantage of many field observations and available data in literature to explicitly simulate the influence of both lateral and vertical PFPs in humid tropical environments and proposed different model structures on the basis of physical reasoning [Savenije, 2009].

The objective of this study was to test the hypothesis that PFPs are responsible for the observed hydrological response of steep and saprolitic catchments in the humid tropics. This hypothesis was tested by proposing a physically based distributed model with four different model structures of increasing complexity informed by hydrological, geochemical, and geophysical observations. Most model parameters were estimated from field observations as well as from literature both quantitatively and qualitatively, with

calibration limited only to a few macropore-related parameters, which is different from typical model development in simulating complex hydrological processes where many parameters were calibrated. The available hydrological data and multiple criteria for model performance evaluation were used to identify plausible model structures and examine whether certain model structures without and with PFPs could possibly explain the hydrological behavior observed in a small tropical catchment, using data from the Smithsonian Tropical Research Institute Agua Salud Project in Central Panama. After identifying the plausible model structures, we also performed sensitivity analysis on the influence of the most uncertain parameters on model performance in capturing hydrological response to potentially guide future field characterization efforts to reduce uncertainty of model predictions.

2. Methods

2.1. Study Area and Relevant Research

Bruijnzeel [2004] stated the importance of maintaining a vigorous experimental field program against which model predictions can be compared in order to retain a realistic perspective. The Smithsonian Tropical Research Institute Agua Salud Project is a large-scale hydrological and ecological land use study started in 2007 in catchments located adjacent to and inside the Panamanian Soberania National Park in Central Panama. Studies consistently showed the effects of PFPs in both lateral and vertical directions formed by tropical root channels and worm burrows on infiltration at the hillslope scale within the Panama Canal Watershed.

From field work in a small catchment within the old (>100 years) forest of Soberania National Park, *Niedzialek and Ogden* [2012] noted the existence of well-developed PFPs such as soil pipes formed by decayed tree roots and macropores created by animal burrows. Overland flow was only observed as return flow from large diameter soil pipes, while high infiltration rates were observed and attributed to these vertical and lateral preferential flows downslope [*Hendrickx et al.*, 2005; *Niedzialek*, 2007].

Using 9 cm diameter soil cores, *Hassler et al.* [2011] measured the K_{sat} values in different Agua Salud Project catchments for shallow soil layers and reported the median soil matrix infiltrability as 2 cm h^{-1} at a depth of 6–12 cm in all land uses ranging from >100 year forest to an active cattle pasture. *Ogden et al.* [2014] performed a series of rainfall simulator tests combined with time-lapse Electrical Resistivity Tomography (ERT) to both measure the infiltration capacity of $2 \text{ m} \times 6 \text{ m}$ areas and visualize the subsurface flow of the infiltrated water under the same land uses. The ERT observations showed that the bulk vertical water velocities greatly exceeded the soil matrix infiltrability measurements by *Hassler et al.* [2011].

Geochemical studies conducted in the Agua Salud catchments also demonstrated the role of well-developed PFPs in hydrological processes. *Litt et al.* [2015] used streamflow electrical conductivity and light stable isotopes of water to perform stream hydrograph separation. The differences in hydrograph separations obtained for larger events suggested activation of another flow path. Following *Kurtz et al.* [2011] in Puerto Rico, *Gardner* [2015] used a three-end-member mixing model to identify flow paths and also interpreted the results as an indication of activated lateral flow paths in the upper 15 cm of the soil. *Gardner* [2015] concluded that the most likely origins for such high capacity shallow lateral flow paths are the tree roots and burrowing animals.

In this study, we proposed four model structures informed by these hydrological, geophysical, and geochemical observations to identify plausible role of PFPs in shaping the hydrological response. The four proposed model structures were tested by applying them to simulate the hydrological response of a 6.5 ha secondary succession catchment, which is a subcatchment of the 3000 km^2 Panama Canal Watershed (Figure 1). The secondary succession catchment is covered by an approximately 10 year-old secondary forest. The mean land surface slope is 24.6° , and the mean topographic index is 4.49, which are all of similar order as the other catchments within the Panama Canal Watershed [*Ogden et al.*, 2013]. The climate of Central Panama has distinct wet and dry seasons. The wet season typically begins in May and ends in December and the dry season is from mid-to-late December to April [*Niedzialek and Ogden*, 2012; *Ogden et al.*, 2013]. Soils in the study catchment are Oxisols derived from in situ weathering of basalt and andesite bedrock [*Ogden et al.*, 2013]. More information of the climate, catchment morphology, and geology and soils can be found in *Ogden et al.* [2013].

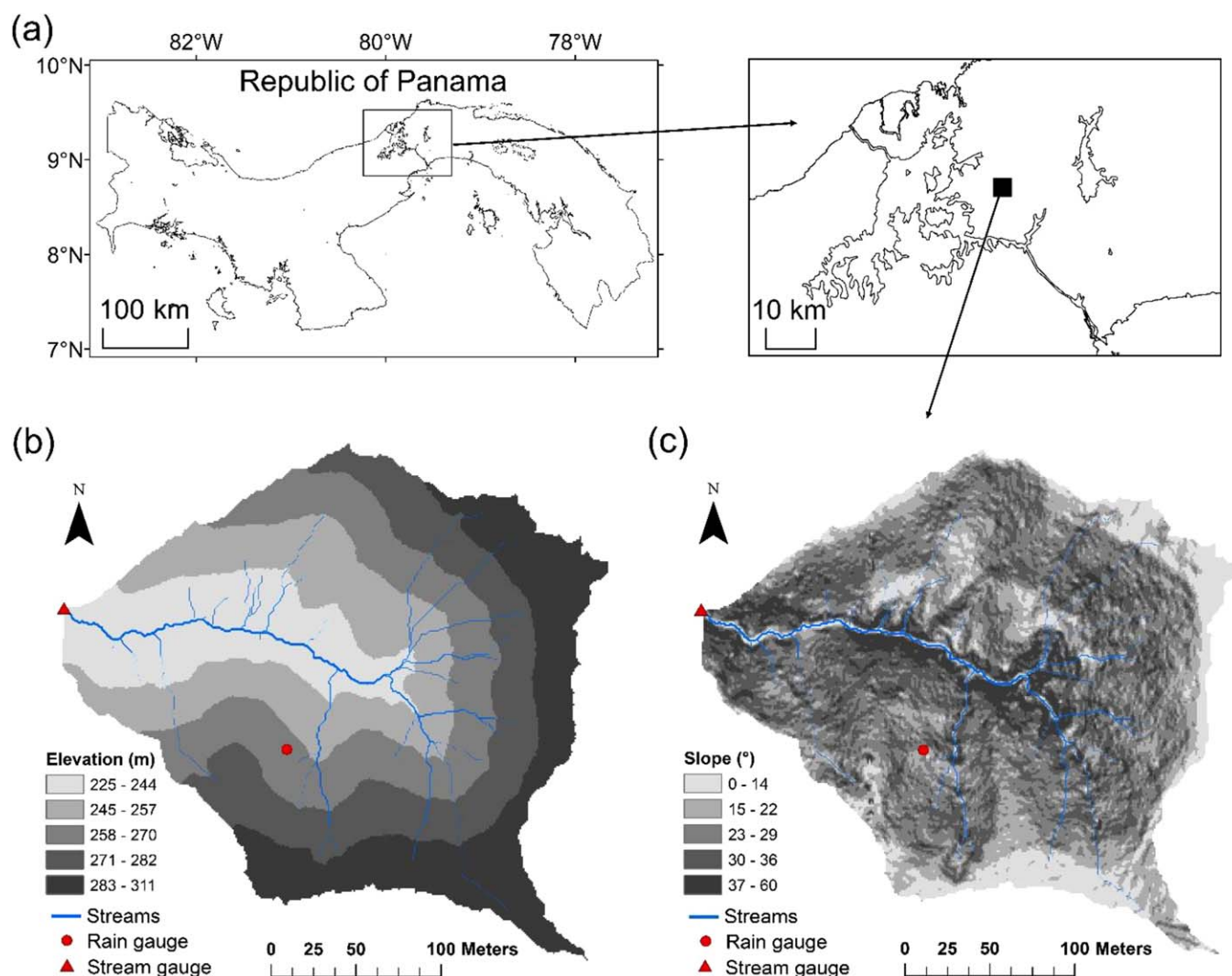


Figure 1. (a) Location, (b) elevation, (c) slope of the 6.5 ha secondary succession catchment which is part of the Smithsonian Tropical Research Institute Agua Salud Project in the Panama Canal Watershed, Panama.

Rainfall data in the study catchment for a period of approximately 8 months (13 February 2012 to 4 October 2012) with 15 min temporal resolution were used to drive the model simulations. This 8 month period begins in the dry season, which extends until late April, and goes about 6 months into the wet season. The rainfall during the simulation period represents average level compared to the general climate of the study area [Niedzialek and Ogden, 2012]. There were 88 major rainfall events during the simulation period, and the cumulative rainfall and runoff was 1590 and 431 mm, respectively.

2.2. Overall Model Framework

We hypothesized that PFPs play an important role in determining the rainfall-rate-dependent runoff generation mechanism in humid tropical catchments. Specifically, we considered a set of near-surface lateral PFPs created by live roots and decayed root channels, commonly known as “soil pipes,” which are dominantly spread out in a lateral direction in tropical environment, and a network of vertical PFPs that are mainly formed by earthworms. Together, the near-surface layer with abundant lateral and vertical PFPs was referred to as a preferential flow layer, underlain by a saprolite layer and fractured bedrock. Figure 2a shows the schematic representation of the vertical soil profile and hydrologic processes at the hillslope scale. The grid size for the squared cells of the proposed physically based distributed model is 25 m. The overall formulation consisted of six components:

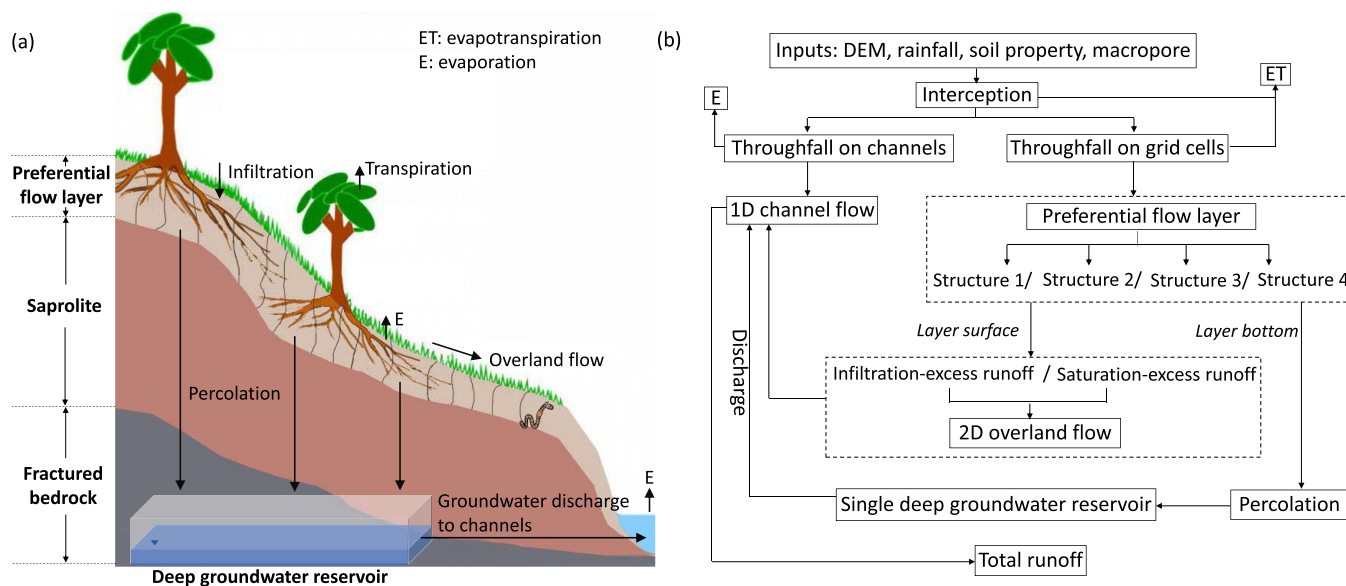


Figure 2. (a) Schematic representation of the hydrologic processes at hillslope scale and (b) flow chart of the common model components and overall computational process in the formulation. Details of the four preferential flow layer structures indicated in Figure 2b are shown in Figure 3.

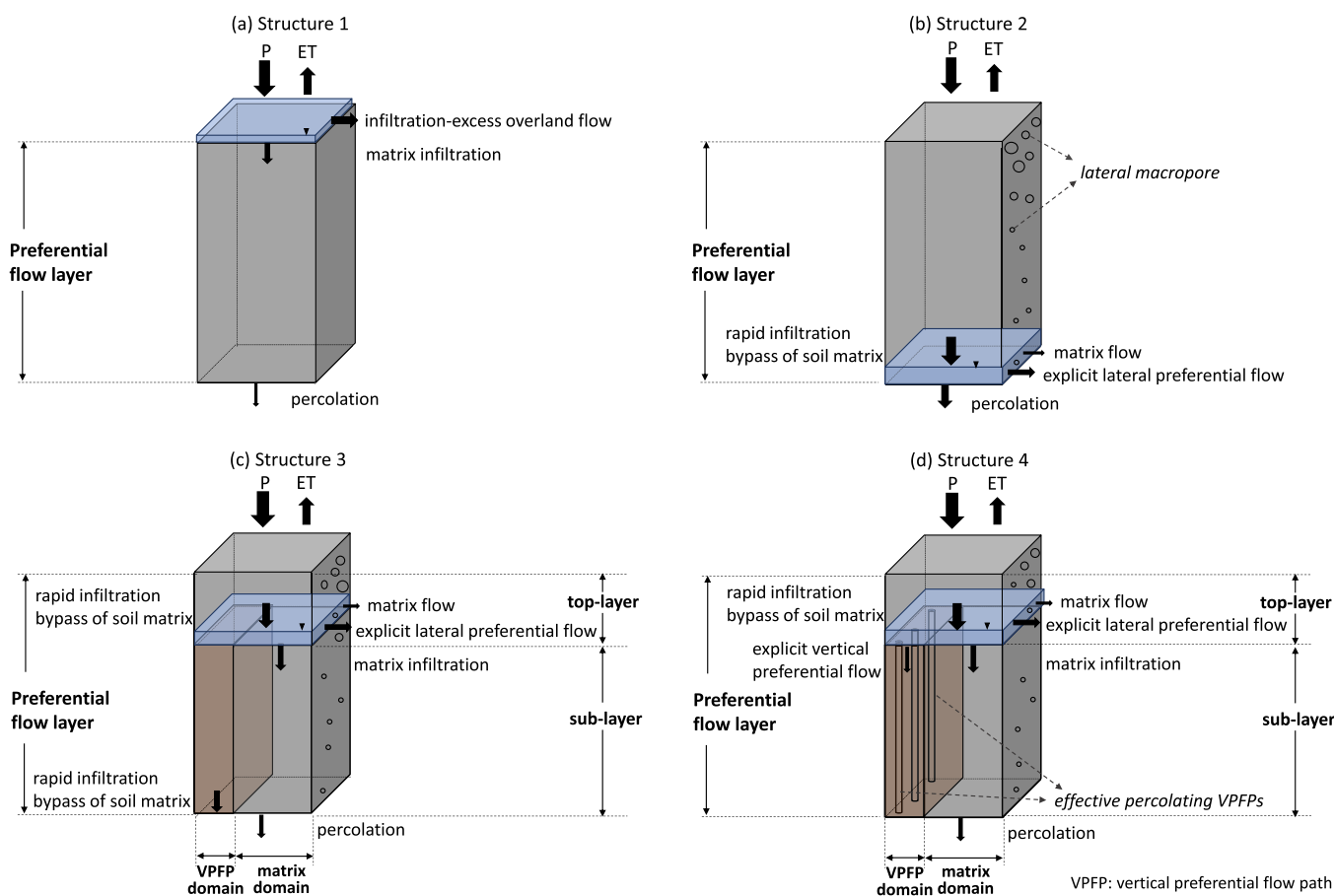


Figure 3. Conceptual diagram of tested model structures for one grid cell: (a) Structure 1, (b) Structure 2, (c) Structure 3, and (d) Structure 4. More detailed descriptions of the water flow paths in the preferential flow layer shown in the figure are given in Table 3.

1. different configurations and water flow algorithms including flows in both the soil matrix and PFPs in the preferential flow layer for each grid cell, which are the main features of this study and will be described in detail later,
2. an evapotranspiration (ET) calculation algorithm extracting available water starting from the surface to the bottom of the preferential flow layer for each grid cell with the constraint that actual ET is less than or equal to the potential ET demand, depending on soil moisture availability,
3. a DEM-based grid cell network for 2-D overland flow routing,
4. a DEM-based channel element network for 1-D channel routing,
5. a percolation calculation as the link between the preferential flow layer and the underlying saprolite soil for each grid cell with the constraint that the recharge rate to deep groundwater is smaller than the maximum percolation rate, and
6. a simple single linear reservoir for simulating deep groundwater discharges to streams.

In component (5), the maximum percolation rate at the preferential flow layer bottom was estimated according to the vertical hydraulic conductivity of the saprolite layer below. In component (6), the groundwater discharge from the linear reservoir was further distributed to each channel element of the stream network based on the topographic index of the element divided by the sum of the topographic indices along the entire stream channel network. The method enables the channel elements at lower elevation with higher topographic index to receive more water than the ones at higher elevation.

A flow chart of the model components common to all tested formulations and an overall computational process is shown in Figure 2b. Table 1 gives the details of these hydrological components, equations, and the involved variables and parameters. Table 2 lists the input parameter values for the overall model formulation and their references. All of these parameters were qualitatively or semiquantitatively estimated from field observations such as ERT experiments, isotope and geochemical tracer tests or from the literature. The Manning roughness coefficients for the overland and channel flow were assigned to be 0.07 and 0.03, respectively [Arcement and Schneider, 1989]. Interception of rainfall by this 10 year-old secondary forest was

Table 1. (a) Common Hydrological Components and Equations and (B) Involved Variables and Parameters Incorporated in the Overall Model Framework

(a) Hydrological Components	Equations	References
Evapotranspiration	$AET = \min (PET \times \Delta t, AW)$	This study
Overland flow	$q_x = \frac{\sqrt{S_x}}{n_1} \cdot h^{\frac{5}{3}}, q_y = \frac{\sqrt{S_y}}{n_1} \cdot h^{\frac{5}{3}}$	Downer and Ogden [2004] and Niedzialek and Ogden [2004]
Channel flow	$Q_c = \frac{1}{n_2} AR_c^{\frac{2}{3}} S_f^{\frac{1}{2}}$	Downer and Ogden [2004]
Percolation	$h_p = \min (f_{\max} \cdot \Delta t, h_v)$	This study
Deep groundwater reservoir discharge	$q_r = K \cdot S, q_c = q_r \cdot \frac{T_c}{\sum_{j=1}^n T_{cj}}$	Beven and Kirkby [1979] and Moore [1997]
(b) Variable and Parameter	Description	Units
AET	Actual evapotranspiration depth for each grid cell	L
PET	Potential evapotranspiration demand	$L T^{-1}$
Δt	Current time step	T
AW	Available water depth for each grid cell	L
q_x, q_y	Unit discharges in x and y directions for overland flow	$L^2 T^{-1}$
S_{fx}, S_{fy}	Friction slopes in x and y directions for overland flow	
n_1	Manning roughness coefficient for overland flow	
h	Surface water depth	L
Q_c	Discharge in or out of each channel element	$L^3 T^{-1}$
n_2	Manning roughness coefficient for channel routing	
A	Channel cross-sectional area	L^2
R_c	Channel hydraulic radius	L
S_f	Channel friction slope	
h_p	Actual percolation depth for each grid cell	L
$f_{\max}$	Maximum percolation rate	$L T^{-1}$
h_v	Available water depth for each grid cell	L
q_r	Total discharge to all the channel elements from the deep groundwater reservoir	$L T^{-1}$
K	Linear reservoir recession constant	T^{-1}
S	Current groundwater reservoir storage	L
q_c	Discharge allocation to each channel element	$L T^{-1}$
n_c	Total number of channel elements	
T_c	Topographic index for each channel element	

Table 2. Input Parameter Values Used in Model Simulations Relevant to Overall Model Framework Input Parameters

Input Parameters	Descriptions	Values	References
n_1	Manning roughness coefficient for overland flow routing	0.07	<i>Arcement and Schneider</i> [1989]
n_2	Manning roughness coefficient for channel routing	0.03	<i>Arcement and Schneider</i> [1989]
I	Interception (%)	13.0	<i>Zimmermann et al.</i> [2013]
PET	Potential evapotranspiration demand (mm d^{-1})	4.5	<i>Schellekens et al.</i> [2000] and <i>Niedzialek and Ogden</i> [2012]
ϕ	Porosity	0.6	B. Harrison (personal communication, 2016)
D	Preferential flow layer thickness (m)	2.0	B. Harrison (personal communication, 2016)
K_{sat}	Matrix saturated hydraulic conductivity (cm h^{-1})	2.0	<i>Hassler et al.</i> [2011]
f_{max}	Maximum percolation rate (cm h^{-1})	2.0	F. L. Ogden (personal communication, 2016)
K	Linear reservoir recession constant (h^{-1})	0.0012	Model warm-up
S_{ini}	Initial reservoir storage (mm)	140	Model warm-up
θ	Initial matrix water content	0.35	Model warm-up

measured to be 13% [Zimmermann et al., 2013]. Following Schellekens et al. [2000] and Niedzialek and Ogden [2012], we set potential ET to be 4.5 mm d^{-1} . Based on the studies of soil properties in nearby test pits, the soil porosity was estimated to be 0.6, and the thickness of the preferential flow layer was estimated to be around 2 m under forest land use (B. Harrison, personal communication, 2016). The saturated hydraulic conductivity of the soil matrix was previously measured as 2 cm h^{-1} [Hassler et al., 2011], and this value was also used to represent the maximum percolation rate at the bottom of the preferential flow layer. The soil matrix water content and groundwater reservoir storage were initialized using a warm-up period of four wet-season months [Aubert et al., 2003].

2.3. Model Structures

This study focuses on the vertical infiltration and lateral subsurface flows in the preferential flow layer. We proposed four model structures with different preferential flow layer configurations shown in Figure 3 with varying levels of complexity that were informed by hydrological, geochemical, and geophysical observations. Table 3 summarizes the model structure options and water flow pathways in the preferential flow layer considered in this study, and Table 4 gives the options used in the four tested model structures. The

Table 3. Summary of Preferential Flow Layer Structure Options and Water Flow Paths Considered in This Study

Options	Names	Descriptions	References
Preferential flow layer structure options	<i>Top-layer architecture</i>		
	A. Rapid vertical infiltration domain	Rapid infiltration bypass of soil matrix in the domain	This study, Structures 3 and 4
	<i>Sub-layer architecture</i>		
	A. Matrix domain	Soil matrix flow in the domain	This study, Structures 3 and 4
	B. VPFP domain	Rapid water flow bypass of soil matrix in the domain	This study, Structures 3 and 4
Preferential flow layer water flow paths options	<i>Vertical infiltration</i>		
	A. Matrix infiltration	GARTO finite water content method	Ogden et al. [2015], Lai et al. [2015], and Talbot and Ogden [2008]
	B. Rapid infiltration bypass of soil matrix	Simply assuming all the water immediately falls into the domain bottom before soil saturation is reached	Kohler et al. [2003]
	C. Explicit vertical preferential flow	Water flows downward through the VPFPs based on Poiseuille's law	Sutera and Skalak [1993]
	<i>Subsurface lateral flow</i>		
	A. Shallow saturated soil matrix flow	Explicit flow routing using fluxes calculated from Darcy's law	Niedzialek and Ogden [2012]
	B. Explicit lateral preferential flow	Water flows laterally through the LPFPs based on Poiseuille's law	Sutera and Skalak [1993]

Table 4. Model Options Used in the Four Model Structures for the Preferential Flow Layer^a

Options	Structure 1	Structure 2	Structure 3	Structure 4
Top-layer architecture			A	A
Sub-layer architecture			A + B	A + B
Vertical infiltration	A	B	A + B	A + C
Subsurface lateral flow	A	A + B	A + B	A + B

^aDetailed descriptions for A, B, and C are given in Table 3.

input parameter values and the sources of these values are listed in Table 5. In this section, we describe how the four tested model structures in this study were conceptualized.

2.3.1. Model Structure 1

This structure assumed that Darcian matrix flow is the only type of flow in the unsaturated zone. We used the GARTO scheme [Lai et al., 2015] to simulate soil matrix flow. The GARTO scheme agrees closely with the numerical solution of Richards' equation [Richards, 1931] but is much more computationally efficient with guaranteed convergence and mass conservation. This is a rainfall-rate-dependent infiltration-excess runoff generation mechanism as shown in Figure 3a.

2.3.2. Model Structure 2

LPFPs cause a much larger effect on infiltrability in tropical forests compared to those in temperate regions [Gerke and van Genuchten, 1993]. As a result, the preferential flow layer may potentially absorb almost all the throughfall [Beven and Germann, 1982]. In this model structure, we assumed that all the throughfall immediately moves to the bottom of the preferential flow layer. In this formulation, overland flow can only occur by the saturation-excess mechanism. After the throughfall immediately moves to the bottom of the preferential flow layer, a portion of water may percolate and recharge to deep groundwater reservoir based on the vertical hydraulic conductivity of the saprolitic soil underneath. During periods where the rain rate exceeds that percolation rate, a perched water table is formed at the interface of the preferential flow layer and the saprolitic soil, triggering lateral flow in both the soil matrix and LPFPs as shown in Figure 3b.

The effectiveness of LPFPs in transmitting water downslope depends on their spatial distribution in vertical direction, pore number, pore size, and connectivity [Beven and Germann, 1982; Elsenbeer and Cassel, 1990; Schellekens, 2000]. Since the density of LPFPs typically follows a decreasing trend with depth in the soil profile [Sidle et al., 2001], an exponentially decreasing distribution was applied to describe the vertical distribution of LPFPs. Based on the results reported by Sidle et al. [2001], we used a LPFP number density of 75/m². The LPFP diameter distribution was assumed to be normally distributed with the mean and standard deviation of 20.0 and 9.0 mm, respectively [Sidle et al., 2001]. LPFPs were assumed to be well connected in each

Table 5. Stochastic Preferential Flow Layer Input Parameter Values Used in the Four Tested Model Structures

Input Parameters	Descriptions	Values	Model Structure No. (This Study)	References
d	Top-layer thickness (cm)	20.0	Structures 3 and 4	B. Harrison (personal communication, 2016)
N_l	LPFP number (per m ⁻²)	75	Structures 2–4	Sidle et al. [2001]
μ_l	Mean of LPFP diameter under normal distribution (mm)	20.0	Structures 2–4	Sidle et al. [2001]
δ_l	Standard derivation of LPFP diameter under normal distribution (mm)	9.0	Structures 2–4	Sidle et al. [2001]
p	The fixed percentage of quick water flow in the VPFP domain (%)	0.0–100.0	Structure 3	Calibration
α	VPFP connectivity parameter	1.0–2.0	Structure 4	Calibration
l_v	VPFP length (cm)	20.0–50.0	Structure 4	Capowiez et al. [1998] and Bastardie et al. [2002]
N_v	VPFP number (per m ⁻²)	30–3000	Structure 4	Munyankusi et al. [1994], Zou and Gonzalez [1997] and Lamandé et al. [2011]
μ_v	Mean of VPFP diameter under lognormal distribution (mm)	2.0	Structure 4	Edwards et al. [1988] and Munyankusi et al. [1994]
δ_v	Standard derivation of VPFP diameter under lognormal distribution (mm)	1.5	Structure 4	Edwards et al. [1988] and Munyankusi et al. [1994]

grid cell as field tests clearly showed the long interconnection of lateral flow paths [Noguchi *et al.*, 1999; Sidle *et al.*, 2001; Uchida *et al.*, 2001]. All the LPFP-related parameters are listed in Table 5. Since we assumed that the LPFP formation is restricted to tropical tree roots, all the LPFP parameter were derived from studies of soil pipes formed by root growth and decays.

Latin hypercube sampling [McKay *et al.*, 2000] was then used to generate the actual pore sizes based on the given pore number and the distribution, mean, and standard derivation of the pore diameter. Or [2008] suggested that flow is likely to be poorly described by Richards' equation in pores with diameters greater than 1.2 mm. The volumetric flow rate in one lateral macropore was calculated by applying the Poiseuille's law in pipes [Sutera and Skalak, 1993]. The total lateral macropore flow through one cell to the neighboring cell was computed as

$$Q_l = \sum_{k=1}^M \left(\frac{(\Delta P_{lk} + \rho g \Delta z_{lk}) \pi}{8 \mu L_{lk}} \cdot R_{lk}^4 \right) \quad (1)$$

where Q_l is the lateral macropore discharge ($L^3 T^{-1}$), ΔP_l ($M L^{-1} T^{-2}$), and Δz_l (L) are the pressure and elevation differences for the lateral macropore, respectively, ρ is the water density ($M L^{-3}$), g is the gravitational acceleration ($L T^{-2}$), μ is the water viscosity ($M L^{-1} T^{-1}$), R_l is the lateral macropore radius (L), L_l is the lateral macropore length (L), and M is the total number of the lateral macropores below the perched water table for each grid cell.

2.3.3. Model Structure 3

Based on the interpretation of geochemistry data collected in this region [Gardner, 2015; Litt *et al.*, 2015] and hydrological observations by others in tropical catchments [Schellekens *et al.*, 2004; Kurtz *et al.*, 2011] that a quick near-surface downslope flow path was active in a shallow soil layer, we divided the preferential flow layer into two horizontal layers as shown in Figure 3c. The upper shallow soil layer with a specified thickness is referred to as top-layer with both abundant LPFPs and VPFPs resulting from high density of roots and earthworm burrow activities [Bonell, 1993], and the bottom, less permeable portion with relatively fewer LPFPs is called the sub-layer.

The top-layer thickness was estimated as 20 cm (B. Harrison, personal communication, 2016), which is of similar order as other studies have observed [Bachmair *et al.*, 2009; Zwartendijk *et al.*, 2017]. Similar to Model Structure 2, this model structure also assumed rapid vertical infiltration so all throughfall bypasses the soil matrix and immediately reaches the bottom of the top-layer and may create a perched water table if the rainfall rate exceeds the percolation capacity of the sub-layer. When this occurs then both the soil matrix and LPFPs below the perched water table in the top-layer will channel water laterally down slope.

Time-lapse ERT observations during rainfall simulation experiments by Ogden *et al.* [2014] visualized the subsurface flow of the infiltrated water, and observed bulk vertical water velocities as high as 20–30 cm h^{-1} , which exceeds soil matrix infiltrability measured by Hassler *et al.* [2011] of 2 cm h^{-1} . Therefore, a two-domain flow algorithm was applied to allow some infiltrated water to bypass the soil matrix in the sub-layer. This was accomplished by adding a conceptual VPFP domain using a parameter representing a fixed fraction of top-layer water that enters the VPFP domain at each time step. Water infiltration occurs in both the VPFPs and the soil matrix but water in the VPFPs reaches the sub-layer bottom very quickly and goes below the preferential flow layer to become groundwater recharge. In the matrix domain, downward water movement was simulated using the GARTO method [Lai *et al.*, 2015]. Water from the VPFP domain was assumed to not interact with the matrix, and LPFPs and VPFPs were treated as disconnected.

2.3.4. Model Structure 4

In this structure, VPFPs were calculated explicitly using a physically based flow model and stochastic parameters. This detailed formulation was proposed to explore the main physical processes behind the fraction of water that reaches the sub-layer bottom very quickly by the VPFPs in Model Structure 3. We divided the preferential flow layer into two horizontal layers (top-layer and sub-layer) and two vertical domains (matrix domain and VPFP domain) as shown in Figure 3d. The top-layer has the same formulation as Structure 3. However, in the VPFP domain in the sub-layer, the VPFPs were assumed to be generated by earthworms.

The required VPFP inputs include the number density, size distribution, length, and the probability of connectivity in the sub-layer as given in Table 5. Together, these parameters determine the fraction of water that bypasses the soil matrix in the preferential flow layer to become groundwater recharge that sustains base flow. Given the lack of field characterizations on the site about the vertical macropore features, all of

these parameters used to describe the VPFPs were taken from relevant quantitative analysis of earthworm burrows from the literature.

VPFPs are primarily circular but with varying size [Beven and Germann, 1982]. Data from the literature indicated that earthworm burrows are predominately 1–10 mm in diameter [Edwards *et al.*, 1988; Munyankusi *et al.*, 1994; Bastardie *et al.*, 2005], but the size distribution is highly skewed [Edwards *et al.*, 1988; Meng *et al.*, 2016]. Their size distribution can be described using either a normal or log-normal distribution [Edwards *et al.*, 1988; Munyankusi *et al.*, 1994]. In this study, the arithmetic mean and the standard deviation of the VPFP diameter were estimated as 2.0 and 1.5 mm using a lognormal distribution following Edwards *et al.* [1988] and Munyankusi *et al.* [1994].

Results derived from macropore reconstruction studies showed that earthworm burrows are primarily vertically oriented [Aubertin, 1971; Langmaack *et al.*, 1999]. However, Munyankusi *et al.* [1994] demonstrated that assuming all visible macropores on the surface are continuous to deeper soil depth is not appropriate. In this model structure, we applied the concept of VPFP connectivity. In order to explore the VPFP connectivity, the sub-layer was further divided into slices with the thickness of each slice being approximately the mean length of the vertical macropores. The VPFPs could then be assumed to be continuous in each slice.

After dividing the sub-layer into slices with continuous VPFPs, we used a function to determine the probability that VPFPs in two different slices are connected. We assumed that the probability of connection is proportional to the total effective pore area and number in the individual slices. The effective pore area per grid cell for slice i , A_{pi} , was calculated using equation (2):

$$A_{pi} = m_i \cdot \pi \cdot [E(\bar{r}_i)]^2 = m_i \cdot \pi \cdot e^{2(\mu_i + \delta_i^2)} \quad (2)$$

where i is the slice index, m_i is the total macropore number in slice i , $E(\bar{r}_i)$ is the effective radius of the macropores in slice i , and μ_i and δ_i^2 are the arithmetic mean and the variance of the macropore diameter in slice i , respectively.

Vertical macropores that are fully penetrating below the preferential flow layer are called “effective percolating VPFPs,” because they are the macropores that allow a portion of throughfall to quickly bypass the soil matrix and percolates the preferential flow layer bottom. The number of effective percolating VPFPs per grid cell, N , was then calculated using equation (3):

$$N = \left(\frac{\alpha_1^2 A_{p1} m_1}{A_c} \right) \cdot \left(\frac{\alpha_2^2 A_{p2} m_2}{A_c} \right) \cdots \left(\frac{\alpha_n^2 A_{pn} m_n}{A_c} \right) = \frac{1}{A_c^n} \left(\prod_{i=1}^n \alpha_i \right)^2 \cdot \left(\prod_{i=1}^n A_{pi} \right) \cdot \left(\prod_{i=1}^n m_i \right) \quad (3)$$

where A_c is the area of each cell, and n is the total number of slices.

In equation (3), the parameter α_i was used to define how macropores in adjacent slices are considered to be connected with the range of $1 \leq \alpha_i < 2$. It represents the dimensionless distance relative to the radius of the tubular macropores between the centers of the pores from the adjacent slices as shown in Figure 4. If α is equal to 1, it means that the pores are considered as connected with each other only when a macropore on the top slice exactly overlaps the one in the bottom slice. On the other hand, α being close to 2 indicates that the two macropores are considered to be connected as long as they have any overlap. Obviously, a larger α_i value means increased connectivity.

Latin hypercube sampling [McKay *et al.*, 2000] was also applied to generate the actual VPFP sizes. The flow rate in the VPFPs was computed by applying the Poiseuille's law shown in equation (1), where in this case M is the number of effective percolating VPFPs.

2.4. Model Assessment

The evaluation of each model structure included analysis of total runoff, runoff peaks, and base flow. Both low and high flows in addition to total runoff were evaluated to address the model ability in capturing different flow variability. It is first necessary to outline some basic principles of hypothesis testing by setting limits-of-acceptability [Beven, 2010]. In order for a model to be considered as plausible, we used the criteria that the model structure needs to have an absolute runoff Volume Error (VE) smaller than 15%, and a Nash-Sutcliffe efficiency (NSE) for the actual discharge values, an NSE for the squares of the discharge data (NSE_{sq}), and an NSE for the logarithms of the discharge data (NSE_{log}) all greater than 0.5. Positive and

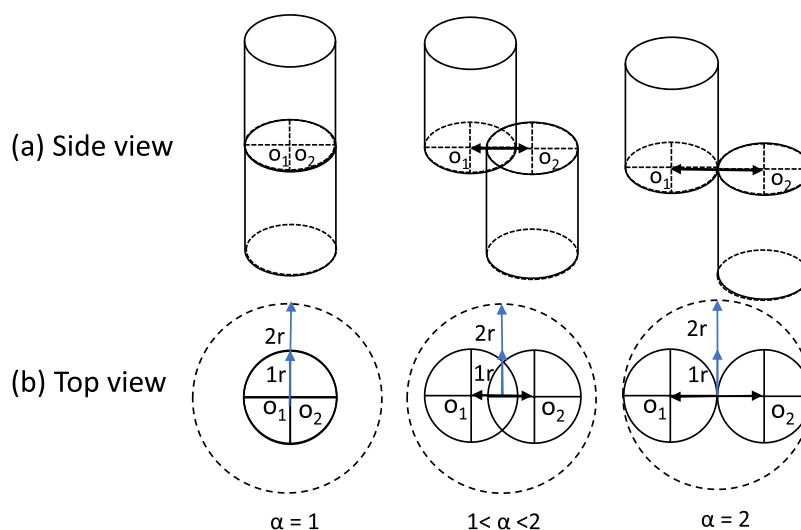


Figure 4. Schematic illustration of the α parameter representing vertical macropore connectivity in equation (3), (a) side view and (b) top view.

negative VE values indicate overestimation and underestimation of total runoff compared to the observations, respectively. In this study, VE together with NSE was used to evaluate the overall model performance. Base flows are typically 1 or more orders of magnitude smaller than the peak discharges and are often poorly reproduced by hydrological models [Staudinger *et al.*, 2011]. The NSE_{log} gives equal weight to peak flow and base flow and can be better used in measuring a model's capability in reproducing base flow, while the NSE_{sq} is better at measuring the simulation performance of runoff peaks. Therefore, the goodness-of-fit value of 0.5 for all three NSEs was used in this study because the purpose of the study was to evaluate model structures' performance in simulating the total runoff, stormflows and base flows at the same time [Samuel *et al.*, 2012].

2.5. Model Sensitivity Analysis

The use of field-observed or literature-derived parameter values allows us to avoid general calibration. Most notably, the parameters used to characterize macropore statistics including the number of macropores, the mean, and the standard deviation of the macropore diameter for both LPFP and VPFP were estimated from the data reported from studies conducted on soils elsewhere. Meng *et al.* [2016] showed that the formation of macropores differs between forest types as a function of their level of diversity. Obviously, these estimated parameters represent the most uncertain aspect of the model testing in this study.

Therefore, we also performed simple sensitivity analysis to examine the influence of the most uncertain parameters on the model structure performance after identifying the plausible model structures. We analyzed the influence of the macropore number, the mean and the standard deviation of the macropore diameter on the model performance criteria such as VE and NSEs. In addition, we also investigated the influence of vertical distribution of LPFPs and size distribution of PFPs on the model performance.

Sensitivity analysis of the LPFP-related and VPFP-related parameters was designed to evaluate the most sensitive parameters in capturing the runoff generation mechanisms. While the model allows better quantification of water partition in the humid tropical catchment, further examination of the complex interplay of LPFPs and VPFPs and their effects in shaping hydrological response from sensitivity analysis is crucial for guiding future field characterization efforts to reduce uncertainty of model predictions.

3. Results and Discussion

The measured and simulated discharges from the four tested model structures are plotted hourly over the simulated period (Figure 5). The scatterplots of both the observed and simulated discharges on a logarithmic scale and the performance measures of VE and NSEs for the four model structures are given in Figure 6. Water balance components for each tested structure are listed in Table 6. There were large differences in

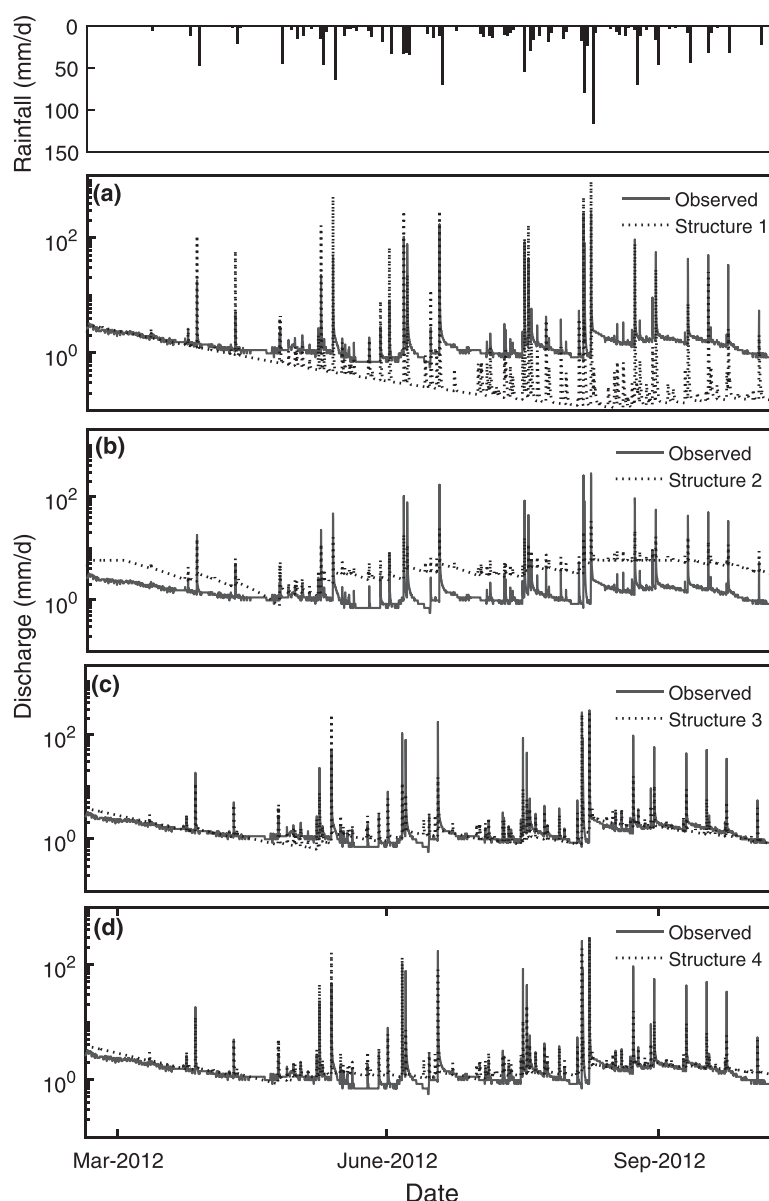


Figure 5. Comparison between the observed and simulated discharge for the period 13 February 2012 to 4 October 2012 for the four tested model structures: (a) Structure 1, (b) Structure 2, (c) Structure 3, and (d) Structure 4.

total runoff, soil moisture storage, and deep groundwater storage changes between the four tested structures, while relatively small differences in actual evapotranspiration, interception, and surface water storage changes were observed.

Results showed that only two model structures with both water flow in the matrix and rapid flow in the PFPs (Structure 3 and Structure 4) passed the criteria in representing catchment hydrological processes and well captured stream discharge patterns. These two model structures produced absolute VEs smaller than 15%, and NSEs for actual, squared, and log-transformed discharge data were all greater than 0.5. The dramatic increase in NSE was particularly obvious from negative values for Structures 1 and 2 to above 0.5 for Structures 3 and 4. It should be noted, however, that Structure 4 with explicit VPFPs had lowest VE and highest NSEs. From the results of performance criteria for the four tested structures, we can see that incorporating both lateral and vertical PFPs in the preferential flow layer by using parameters estimated from the field observations and literature significantly improved model performance in capturing hydrological behaviors based on the objective measures for total runoff, runoff peaks, and base flow.

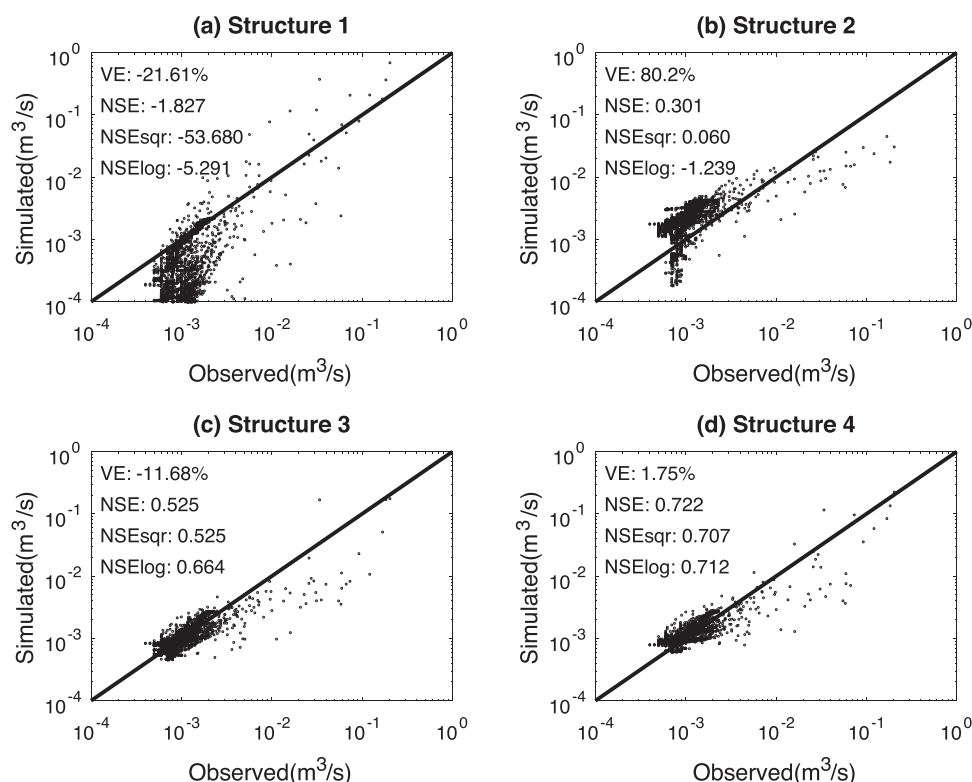


Figure 6. Scatterplots of the simulated discharge and observed discharge and performance statistics (runoff Volume Error VE, Nash-Sutcliffe efficiency of actual discharge values NSE , Nash-Sutcliffe efficiency for the squares of the discharge data NSE_{sqr} , and the Nash-Sutcliffe efficiency for the logarithms of the discharge data NSE_{log}) for the period 13 February 2012 to 4 October 2012 for the four tested model structures: (a) Structure 1, (b) Structure 2, (c) Structure 3, and (d) Structure 4. Notes: The equations used to calculate the performance

$$\text{statistics are } VE = \frac{R^{sim} - R^{obs}}{R^{obs}} \times 100\%, \quad NSE = 1 - \frac{\sum_{i=1}^n (Q_i^{sim} - Q_i^{obs})^2}{\sum_{i=1}^n (Q_i^{obs} - \bar{Q}^{obs})^2}, \quad NSE_{sqr} = 1 - \frac{\sum_{i=1}^n ((Q_i^{sim})^2 - (Q_i^{obs})^2)^2}{\sum_{i=1}^n ((Q_i^{obs})^2 - (\bar{Q}^{obs})^2)^2}, \quad NSE_{log} = 1 - \frac{\sum_{i=1}^n (\log Q_i^{sim} - \log Q_i^{obs})^2}{\sum_{i=1}^n (\log Q_i^{obs} - \log \bar{Q}^{obs})^2}$$

where R^{obs} and R^{sim} are observed and simulated values of runoff, respectively; Q^{obs} and Q^{sim} are the observed and simulated values of discharges, respectively; $\bar{Q}^{obs}$ is the average of Q^{obs} ; and n is the number of data points.

3.1. Model Structure Evaluations

3.1.1. Model Structure 1

Structure 1 produced less total runoff and unsatisfactory overall performance as demonstrated by negative VE (−21.61%) and NSE (−1.827) (Figure 6). Both negative NSE_{sqr} (−53.68) and NSE_{log} (−5.291) values also indicated its unsatisfactory performance in capturing high and low flows. For runoff peaks, due to the relatively low saturated hydraulic conductivity of the soil matrix, the matrix water flow algorithm persistently produced more direct surface flow, resulting in much higher peak flow compared to observations (Figure 5).

Table 6. Water Balance Components (Total Runoff R , Actual Evapotranspiration AET , Interception I , Soil Moisture Storage Changes ΔS_{soil} , Deep Groundwater Storage Changes ΔS_{gw} , and Surface Water Storage Changes ΔS_{sur}) for the Four Tested Model Structures^a

Water Balance Components	Structure 1	Structure 2	Structure 3	Structure 4
R (mm)	336.94	776.85	380.76	438.64
AET (mm)	848.96	892.34	900.49	875.32
I (mm)	206.33	206.22	206.66	206.55
ΔS_{soil} (mm)	305.45	−252.86	207.62	157.33
ΔS_{gw} (mm)	−125.57	−35.93	−108.34	−93.63
ΔS_{sur} (mm)	−0.55	−0.32	−0.33	−0.26

^aAll the water balance component depth values are relative to the entire catchment area. Total precipitation was 1590 mm and total observed runoff was 431 mm for the simulation period. A positive sign in water storage changes means water gain in the system, while negative sign indicates water loss during the simulation period.

The overestimation of peak runoff rates suggested that subsurface flow should be more important than overland flow for runoff generation mechanism in this tropical secondary forest. Although the K_{sat} results by Hassler *et al.* [2011] implied that overland flow is likely in all land covers, their data were concerned with small core permeametry for shallow soil layers only and gave actual median values relative to median and 95% rainfall intensity values. The model results in this study were consistent with the field rainfall simulator experiments conducted by Ogden *et al.* [2014] that rare overland flow was observed in forest catchments in Panama.

Moreover, the lack of quick flow in PFPs in this structure resulted in a significant delay before wetting front reaches the bottom of the preferential flow layer. Much water was stored in the preferential flow layer as soil moisture and little water was supplied from the preferential flow layer to deep groundwater (Table 6). As a result, base flow was significantly underestimated. The results demonstrated that a matrix flow only algorithm does not represent soil water flows in this forest catchment.

3.1.2. Model Structure 2

The algorithm tested in this model structure, which incorporates LPFPs, produced much less infiltration-excess overland flow and can better simulate runoff efficiencies typically demonstrated in Ogden *et al.* [2013] compared to the matrix-only infiltration in Structure 1 with higher NSE (0.301) and NSE_{sqr} (0.06), indicating the key role of LPFPs in delivering water quickly to the stream.

The results also indicated that the assumption that all the throughfall immediately reaches the preferential flow layer bottom is not true because it led to unrealistically high vertical flow in the preferential flow layer and high percolation rates to deep groundwater, resulting in excessive base flow and total runoff (Figure 5) with unsatisfactory NSE_{log} (−1.239) and VE (80.2%). The fact that this model structure significantly overestimated base flow also meant that soil moisture storage was significantly underestimated (Table 6) because of the increased percolation. The results illustrated that while the preferential flow layer in the tropical forests is highly permeable, not all the throughfall can reach the bottom immediately.

The poor performance of both Model Structures 1 and 2 indicated that their formulations are implausible, because they either severely underestimate or overestimate groundwater recharge. Since we have used the parameters estimated from the field observations or from the literature with no calibration in these two structures, we can conclude that this is a model structural error not parameter error. Other more realistic representations of the hydrological processes in the preferential flow layer are required.

3.1.3. Model Structure 3

Structure 3 assumed that flows through the VPFPs and the soil matrix in the sub-layer are both important in determining the amount of vertical infiltration and in apportioning water that becomes available for LPFP activation in the top-layer. This model structure assumed a constant fraction of the throughfall bypasses the soil matrix and travels through the VPFPs. This bypass fraction was calibrated, and a value of about 8% was found to produce reasonable agreement with the observed discharge series (Figure 5). Total runoff was more satisfactorily reproduced as illustrated by lower VE (−11.68%) and higher NSE (0.525). The division of the preferential flow layer into two layers and two domains, which was supported by field-based understanding of the hydrological processes, as demonstrated by simulated rainfall simulator experiments with ERT and tracer observations in the study area, can also better simulate both runoff peaks and base flow patterns compared to the previous two model structures with significantly higher NSE_{sqr} (0.525) and NSE_{log} (0.664). The results supported the idea that preferential flow layer configurations in general and VPFPs in particular strongly dictate groundwater recharge mechanisms in tropical catchments [Cuthbert and Tindimugaya, 2010].

3.1.4. Model Structure 4

This model structure was formulated to allow dynamic allocation of the portion of throughfall that bypasses the soil matrix and quickly reaches the bottom of preferential flow layer by introducing the notion that VPFPs should be explicitly simulated and their connectivity is important. The creation of a model structure that allows specification of the connectivity of vertical macropores using parameters from related studies on earthworm burrows helped us identify the number of effective percolating VPFPs that fully penetrate the preferential flow layer.

In the secondary succession catchment, the VPFP domain was divided into five slices of varying thickness based on the VPFP length range given in Table 5. We used a value of 1.5 for α for all slices, which was calibrated, in calculating the VPFP connectivity in equation (3). Based on this α value, the effective percolating VPFP number for the entire secondary succession catchment calculated by equation (3) was 2277.

These effective percolating VPFPs transmitted a sufficient portion of throughfall to quickly leave the preferential flow layer and supplied sufficient recharge to the deep groundwater, allowing the model to better capture observed base flow behaviors with highest NSE_{log} (0.712). The structure with explicit VPFPs using a number and size distribution can also better capture both total runoff and runoff peaks with lowest VE (1.75%), highest NSE (0.722) and NSE_{sqr} (0.707), indicating the role of VPFPs in regulating high flow production through their control on determining the water partition between the top-layer and the sub-layer in the preferential flow layer. The better performance of Structure 4 compared to Structure 3 with respect to multiple performance criteria taking into account the total runoff, high and low flows suggested that Structure 4 with explicit VPFPs is more capable of simulating important hydrological processes and flow variability in the tropical forest catchment.

3.2. Sensitivity Analysis

The VE, NSE, and storm flow as a percentage of the total runoff (R_s/R) as functions of LPFP number under different vertical distributions with the same diameter values used in Structure 4 are shown in Figure 7. Three vertical distributions, namely uniform, exponentially decreasing, and normal distributions were tested to represent the decrease of LPFP number over soil depth. In general, there was little difference among these three different vertical distributions based on the multiple criteria (Figure 7). The maximum relative differences of VE, NSE, and R_s/R among the three vertical distributions were all less than 10%. The results also indicated that runoff generation behaviors were not sensitive to the LPFP number after it is above a certain threshold, after which R_s/R (Figure 7c) did not change much, while absolute VEs (Figure 7a) were all less than 4% and NSE_{sqr} values (Figure 7b) were all greater than 0.65. The results demonstrated that the existence of LPFP was of crucial importance; but as long as they exist, the actual number of LPFP does not affect the runoff generation behavior to any significant extent.

Since LPFPs are important for simulating peak flows as discussed earlier, in addition to VE, NSE_{sqr} and storm flow as a percentage of the total runoff (R_s/R) were also chosen to analyze the model sensitivity to the LPFP diameter and its distribution (Figure 8). Note that in analyzing the impact of one LPFP diameter parameter we fixed the other LPFP parameters to the same values used in early simulations.

Results showed that the pore size distribution did not significantly affect both the total runoff and storm flow generation (Figure 8). Similar to the sensitivity results of LPFP number, the model structure performance was not sensitive to the mean of the LPFP diameter after it reaches a certain value (Figures 8a–8c). The results also demonstrated that the model structure performance was not sensitive to the standard deviation of the LPFP diameter to any significant extent (Figures 8d–8f) although some fluctuations exist in the relationship between NSE_{sqr} and LPFP standard deviation.

Because the model results already indicated that VPFPs are dominant in controlling groundwater recharge, NSE_{log} and base flow as a percentage of the total runoff (R_b/R) were thus selected in addition to VE to identify the roles of VPFP number and size (Figure 9). Both normal and lognormal distributions were tested since they were commonly reported as the size distribution of VPFPs. Similar to the LPFP sensitivity analysis, we varied one VPFP parameter while fixing the other VPFP parameters to the values used in early simulations.

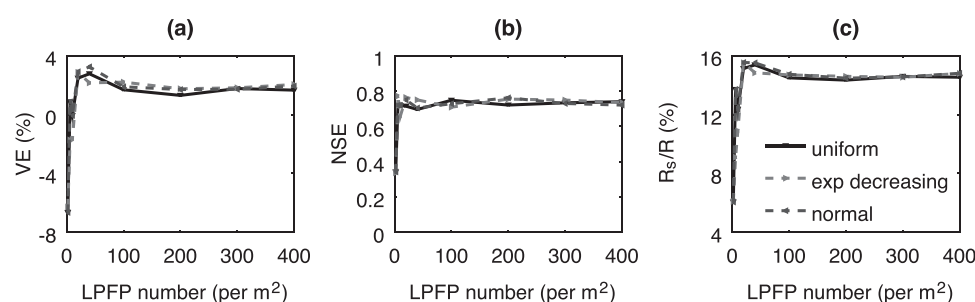


Figure 7. (a) Runoff Volume Error VE, (b) Nash-Sutcliffe efficiency NSE and (c) storm flow as a percentage of the total runoff (R_s/R) variation with the lateral preferential flow path (LPFP) number under three vertical distributions (uniform, exponentially decreasing and normal) along the preferential flow layer profile.

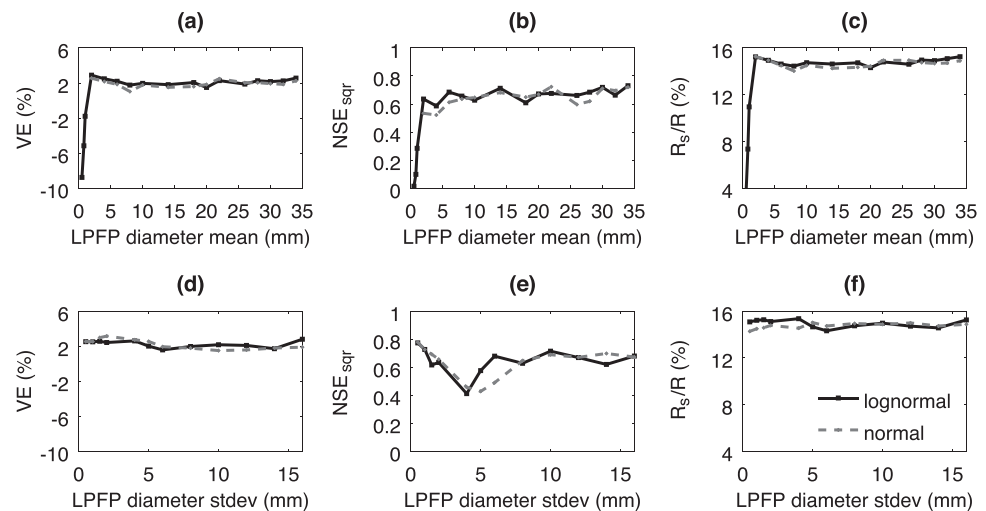


Figure 8. (a) Runoff Volume Error VE , (b) Nash-Sutcliffe efficiency for the squares of the discharge data NSE_{sqr} , and (c) storm flow as a percentage of the total runoff (R_b/R) variation with the mean of the LPFP diameter under normal and lognormal distributions; (d) VE , (e) NSE_{sqr} , and (f) R_b/R variation with the standard derivation (stdev) of the LPFP diameter under normal and lognormal distributions.

The effective percolating VPFP number, the mean and standard derivation of the VPFP diameter were all important in determining the overall runoff and base flow generation, seen from Figures 9c, 9f, and 9i that the R_b/R values increased accordingly with the increase in both the VPFP number and size. The fact that all the R_b/R values were larger than 70% indicated the large contribution of groundwater storage to stream-flow, which was consistent with the results from other studies in tropical catchments [Birkel *et al.*, 2012;

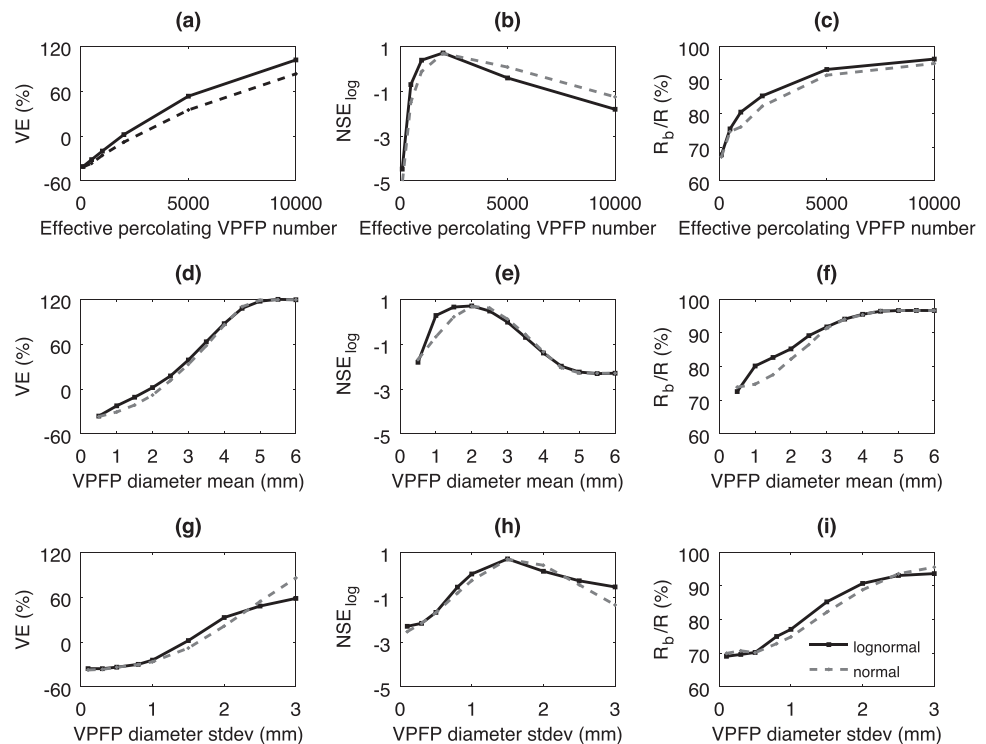


Figure 9. (a) Runoff Volume Error VE , (b) Nash-Sutcliffe efficiency for the logarithms of the discharge data NSE_{sqr} , and (c) base flow as a percentage of the total runoff (R_b/R) variation with the vertical preferential flow path (VPFP) number under normal and lognormal distributions for diameter; (d) VE , (e) NSE_{log} , and (f) R_b/R variation with the mean of the VPFP diameter under normal and lognormal distributions; (g) VE , (h) NSE_{log} , and (i) R_b/R variation with the standard derivation (stdev) of the VPFP diameter under normal and lognormal distributions.

Ogden *et al.*, 2013]. Maximum values of NSE_{log} (Figures 9b, 9e, and 9h) were observed in relation to every VPFP parameter, which mainly corresponded to the values we estimated based on field observations and literature as discussed earlier. Deep groundwater recharge through these effective percolating VPFPs in the preferential flow layer was critical to simulate base flow and total runoff since the satisfactory model performance can only be achieved when the effective percolating VPFP number and size are within a certain small range. Little difference due to size distribution was observed in determining hydrological response.

The VPFPs have a more pronounced effect on the simulation results than the LPFPs (Figures 8 and 9), because the activation of LPFPs depends on the flow capacity of VPFPs, but not vice versa. In this regard, Structure 4 provided evidence of how the VPFPs in the preferential flow layer control deep groundwater recharge and regulate peak flow production through their control on the activation of LPFPs.

3.3. Further Remarks About Model Structures

We have rigorously tested four model structures using the available observations from the secondary succession catchment in the Panama Canal Watershed and found that the two model structures with formulations that incorporate both PFPs and matrix flow are plausible. The reasons for differences in overall performance among different model structures are attributable to the catchment characteristics such as preferential flow layer structure and soil properties. Besides these relatively static variables, dynamic factors such as land use and land cover [Bruijnzeel, 1989] also tend to heavily dictate the PFP characteristics in the preferential flow layer [Bachmair *et al.*, 2009; Colloff *et al.*, 2010; Zimmermann *et al.*, 2013; Zwartendijk *et al.*, 2017]. For instance, if forests are removed, the biological intensity of root growth and animal activity are changed in the shift towards grasslands, which will in turn change the relative rates of PFP creation/destruction. As a result, hydrological behaviors will also change accordingly.

The model structures presented in this study are highly flexible to incorporate the characteristics related to PFPs and land use types of different catchments. These formulations can be extended and may be tested in other data-rich rain forest sites such as La Cuenca (Perú), South Creek (Queensland), the Danum (Malaysian Borneo), Bukit Tarek (Peninsular Malaysia), and the Luquillo Experimental Forest (Puerto Rico) in combination with improved representation of macropore architecture through blue dye experiments and other techniques such as time-lapse ERT and geochemical and light stable isotope tracers.

4. Conclusions

The overarching objective of this study was to improve predictive understanding of how the preferential flow paths (PFPs) affect partitioning of rainfall into runoff, soil moisture, and groundwater recharge in the steep, humid, tropical lowland catchments with saprolitic soils. To achieve this objective, we proposed and tested four different physically based distributed model structures ranging from uniform matrix infiltration to three structures of increasing complexity in incorporating PFPs in an attempt to simulate tropical rainforest hydrology, using data from the Agua Salud Project experimental catchment(s) in Central Panama. Based on the literature and our own observations, we assumed that lateral PFPs were mainly created by tree root channels, while vertical PFPs predominantly consisted of earthworm burrows.

Model parameters describing soil matrix hydraulic properties were estimated from published studies of soil infiltrability performed in nearby catchments. Time-lapse Electrical Resistance Tomography observations made during rainfall simulator tests and geochemical data collected in the study area were used to infer water flow pathways associated with vertical and lateral PFPs. Literature values of some model parameters were used when local measurements were not available. This approach allowed us to minimize the calibration to just a few macropore-related parameters. The hydrological data and multiple criteria for model performance evaluation were used to identify the plausible model structures that may explain the role of PFPs. Given the purpose of capturing both stormflows and base flows in addition to total runoff, runoff Volume Error (VE) and three Nash-Sutcliffe model efficiencies (NSEs), calculated using actual, squared, and log-transformed discharge data, were used collectively in this study.

Poor model performance as exhibited by large absolute VE and negative NSEs in Model Structure 1 showed that a physically based description that relies solely on soil matrix properties was not capable of predicting observed flows in a steep, saprolitic, lowland catchment in the humid seasonal tropics, while the severe

overestimation of base flow in Model Structure 2 demonstrated water flows in both the soil matrix and vertical PFPs were equally important and should be both taken into account.

Comparing Model Structure 3 with the first two, however, showed that the division of the preferential flow layer into two layers and two domains, which was supported by available field data and empirical hydrological process knowledge, can reasonably represent catchment hydrological processes and satisfactorily simulate main observed stream discharge patterns, both in terms of low flow and high flow dynamics. The improved performance of Model Structure 4 compared to Structure 3 indicated that explicit simulation of PFPs using a number and size distribution is superior to the other model structures tested and more capable of simulating preferential flow processes in humid tropical catchments.

One of the most interesting findings of these model structure tests was that a small percentage of vertical PFPs that move water downward below the root zone seems to have a dominant effect on groundwater recharge and base flow. The ability of Model Structure 4 to dynamically simulate the effect of vertical PFP control on lateral PFP activation highlights the importance of knowledge of the properties of vertical PFPs in the tropical forest catchment. The vertical PFP number, the mean and the standard deviation of the vertical PFP diameter all play important roles in capturing overall runoff and base flow generation behaviors. Another one was that the shallow lateral PFPs in the upper 10–20 cm of soil seem to be responsible for the main contributions to peak flows, while their number and size distribution were found to be more important than the vertical spatial distribution.

A model with a formulation similar to the tested Model Structure 3 or 4, with parameter values derived from appropriate observations of lateral and vertical PFP characteristics in different land uses may be a valuable tool for assessing the effect of land use management decisions on the hydrological behavior of heterogeneous catchments. Such a model might prove useful for evaluating different land use management scenarios and payments for ecosystem service schemes. We leave this for future work.

Acknowledgments

This work was funded by the U.S. National Science Foundation (NSF) EAR-1360384 WSC-Category 2 Collaborative Research: Planning and Land Management in a Tropical Ecosystem: Complexities of Land use and Hydrology Coupling in the Panama Canal Watershed, and the Smithsonian Tropical Research Institute. Graduate student support for integrated hydrological model testing was partially provided by the Wyoming Center for Environmental Hydrology and Geophysics (WyCEHG) funded by NSF EPS-1208909. We gratefully acknowledge discussions with Christopher Gardner and W. Berry Lyons, Guy Litt, and Bob Stallard regarding interpretations of geochemical tracer observations. We also thank Alexis Mojica for providing the time-lapse electrical resistivity tomography observations, Ed Kempema, Jason Regina, and Juan Carlos Briceno for their assistance with the rainfall simulator tests, and Bob Steinke for his assistance in programming and computer science aspects of the project. The associate model written in C and data used in this paper are available at <https://doi.org/10.15786/M2GW2J>.

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